

CS19611 - MOBILE APPLICATION DEVELOPMENT

PROJECT REPORT

DAILY QUOTES GENERATOR

Submitted by

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in partial fulfilment for the course for the degree

Of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING



**RAJALAKSHMI
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MAY 2025

BONAFIDE CERTIFICATE

Certified that this project report titled “**DAILY QUOTES GENERATOR**” is the bonafide work of **LAKSHANYA D(220701140)**, who carried out the work under my supervision. Certified further that to the best of my knowledge, the work reported herein does not form part of any other thesis or dissertation based on which a degree or award was conferred earlier.

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ABSTRACT

The **Daily Quote Generator Application** is a lightweight, user-centric Android mobile application designed to deliver inspirational, motivational, or thoughtful quotes to users on a daily basis. Developed using **Android Studio**, the app showcases the fundamental principles of native Android development without relying on external libraries or online APIs. The application is built using **Java** for the backend logic and **XML** for the user interface, ensuring a clean and responsive design.

The core functionality revolves around displaying a new quote each time the user interacts with the app, offering a simple yet engaging experience. Quotes are stored locally within the application resources, and a **random selection algorithm** ensures varied and dynamic content on each launch or button press. The user interface consists of a minimalistic two-screen layout: a **Home Screen** welcoming the user and a **Quote Screen** that displays the randomly selected quote with options to navigate or refresh.

Key features include smooth navigation using **Intents**, user feedback via **Toast messages**, and aesthetic design elements managed through **ConstraintLayout**. The app adheres to principles of clean code and modular development, with a focus on enhancing user engagement and delivering quick, meaningful interactions.

The project aims to provide hands-on experience with Android component integration, activity lifecycle management, and resource handling. Future enhancements may include user-customized quote categories, scheduled daily notifications, and social sharing functionalities.

Overall, the **Daily Quote Generator** serves as a foundational project for Android development learners, combining creativity with technical implementation to encourage personal growth through daily inspiration.

ACKNOWLEDGEMENT

Initially we thank the Almighty for being with us through every walk of our life and showering his blessings through the endeavor to put forth this report. Our sincere thanks to our Chairman **Mr. S. Meganathan, B.E, F.I.E.**, our Vice Chairman **Mr. Abhay Shankar Meganathan, B.E., M.S.**, and our respected Chairperson **Dr. (Mrs.) Thangam Meganathan, Ph.D.**, for providing us with the requisite infrastructure and sincere endeavouring in educating us in their premier institution.

Our sincere thanks to **Dr. S. N. Murugesan, M.E., Ph.D.**, our beloved Principal for his kind support and facilities provided to complete our work in time. We express our sincere thanks to our **DR. P. Kumar** Professor and Head of the Department of Computer Science and Engineering for his guidance and encouragement throughout the project work. We convey our sincere thanks to our internal guide and Project Coordinator, **Dr. V. Karthick**, Rajalakshmi Engineering College for his valuable guidance throughout the course of the project.

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CHAPTER 1

1.INTRODUCTION

In today's fast-paced world, a few words of motivation or wisdom can significantly influence a person's mindset and productivity. The **Daily Quote Generator Application** is a simple yet meaningful Android mobile application developed to provide users with random inspirational quotes each day. This project was initiated with the objective of strengthening core Android development skills while building an app that delivers daily positivity through a clean and user-friendly interface.

The application is developed using **Android Studio**, with **Java** for backend logic and **XML** for frontend design. It uses locally stored quotes and selects them at random, ensuring that users receive a new and refreshing quote every time they interact with the app. Designed for offline use and smooth performance, the app focuses on basic Android components like Activities, Intents, and View elements without the use of third-party APIs or libraries.

This project also serves as a hands-on learning experience in modular code development, UI design, and app lifecycle management. It lays the groundwork for more advanced quote-based applications that could include cloud integration, notifications, or sharing features in the future.

CHAPTER 2

2.LITERATURE SURVEY

The development of mobile applications that provide daily quotes or motivational messages has grown significantly in recent years, driven by the increasing need for mental well-being and daily inspiration. Many popular apps in this category, such as Brilliant Quotes, Motivation Daily, and ThinkUp, offer users curated or customizable quotes along with features like reminders, themes, and sharing options. These apps often integrate cloud-based APIs, push notifications, and social media sharing, enhancing user engagement and personalization.

However, many of these advanced applications rely heavily on third-party libraries, databases, or internet connectivity, which can increase complexity and limit offline usability. In contrast, simple quote generator apps focusing on core Android functionalities are ideal for beginners to understand key development concepts like UI design, activity lifecycle, local data handling, and randomization logic.

Academic studies and tutorials on basic Android application development emphasize the importance of mastering fundamental components such as TextView, Button, Intent, SharedPreferences, and simple data storage techniques. Resources such as the Android Developer Guide and online platforms like GeeksforGeeks, Stack Overflow, and Medium blogs provide essential insights into developing lightweight and functional apps using native tools.

This project draws from these foundational resources and fills the gap for a lightweight, offline-capable, and minimalistic quote app that focuses on clarity, efficiency, and educational value. It sets the stage for future expansion into more feature-rich implementations while remaining rooted in core Android development principles.

Mobile applications that deliver motivational content such as daily quotes have become increasingly popular across both Android and iOS platforms. These applications are designed to promote positive thinking, mental well-being, and productivity through the use of curated or user-customized

messages. The demand for such tools has surged, particularly among students, professionals, and individuals seeking mindfulness support.

Several well-known applications already exist in this domain:

- **Brilliant Quotes** offers over 3,600 quotes from 250 authors, categorized by theme. It includes features like bookmarking, notifications, and sharing.
- **Motivation – Daily Quotes** provides inspirational messages with aesthetic themes and notification reminders, supporting user mood tracking and daily challenges.
- **ThinkUp** allows users to record their own affirmations and play them as audio, combining quotes with personal voice for deeper impact.

While these applications are feature-rich and impactful, they rely heavily on internet connectivity, cloud-based APIs, and third-party libraries such as Retrofit, Firebase, or Glide. These dependencies increase complexity and limit accessibility in offline environments.

In contrast, basic applications like the **Daily Quote Generator** aim to provide similar core functionality using only native Android components such as TextView, Button, ArrayList, and local XML/Java code. This not only simplifies development but also ensures offline compatibility, minimal storage usage, and faster response times.

Additionally, academic resources like the Android Developer Documentation, GeeksforGeeks tutorials, and Stack Overflow solutions have been instrumental in guiding the implementation of simple mobile apps without third-party support. These resources recommend beginning with static data (such as a list of quotes in strings.xml or ArrayList) before exploring dynamic options.

CHAPTER 3

3.PROPOSED SYSTEM

The proposed system is a lightweight and standalone Android application named **Daily Quotes Generator**, designed to deliver motivational or insightful quotes to users every day. The application operates using native Android components and does not rely on third-party libraries or internet connectivity. It aims to provide users with a consistent and uplifting experience, while maintaining a clean, user-friendly interface and a modular code structure that is easy to extend and maintain.

3.1 System Overview

The core functionality of the Daily Quotes Generator is to display a randomly selected quote from a predefined collection every time the user opens the application or refreshes the screen. The quotes are stored locally within the application, ensuring offline access and instant response times. The system supports a simple two-screen interface:

- A **Home Screen** displaying a “Quote of the Day” with options to refresh or navigate.
- A **Quote Gallery Screen** where users can explore more quotes or view previously displayed ones.

The app is built using **Java** and **XML** within **Android Studio**. It ensures compatibility with a wide range of devices and Android versions by using standard Android components and best practices.

3.2 Architecture and Design Philosophy

The application follows a **Model-View-Controller (MVC)**-inspired design pattern for separation of concerns:

- **Model:** A local collection of quotes stored as an array or in a local JSON file.
- **View:** XML layout files (activity_main.xml, activity_gallery.xml) for user interface elements.
- **Controller:** Java classes (MainActivity.java) that manage logic, quote selection, and UI interaction.

This design allows for **scalability**—new features such as bookmarking, sharing, notifications, or quote categories can be easily integrated in future versions.

3.3 Key Features and Components

3.3.1 MainActivity (Quote Display Screen)

The MainActivity is the entry point of the application. It randomly selects a quote from the local collection and displays it prominently to the user with styling and animation.

Core functionalities:

- Random quote generation on launch or button press
- Refresh button to generate a new quote
- Navigation to the quote gallery screen
- Background color or theme changes to enhance user engagement

3.3.2 Local Quote Collection

The quotes are stored in a local structure (such as an array or a JSON file in the **assets** folder). This ensures:

- **Offline usability**
- **Fast data access**
- Easy management and expansion of quote data

3.3.3 GalleryActivity (Quote Archive Screen)

An additional activity that displays all available quotes in a scrollable list or card format. This provides users the ability to revisit previously shown quotes.

Features include:

- Vertical RecyclerView or ListView to show multiple quotes

- **Styled cards** with fonts and backgrounds for better readability
- **Optional search or filter functionality**

3.3.4 User Interface

The app uses `ConstraintLayout` and `TextView` components for flexible UI arrangement. Buttons and quote texts are styled with attention to typography, spacing, and accessibility.

3.4 Workflow: From App Launch to Quote Display

1. **App Launch:** The `MainActivity` is initialized.
2. **Quote Retrieval:** A quote is randomly selected from the local collection.
3. **Display:** The selected quote is shown with animations or styling.
4. **User Interaction:** Users can tap the refresh button to see a new quote or navigate to the gallery for more.

3.5 Advantages of the Proposed System

- **Offline Support:** No internet required to use the app.
- **Fast and Lightweight:** Quick launch and minimal resource usage.
- **Modular Design:** Easy to add new features like daily notifications or quote sharing.
- **User Engagement:** Clean UI, randomization, and interactivity encourage repeated use.

CHAPTER 4

4. MODULE DESCRIPTION

The modular design of the application improves readability, scalability, and performance. Each core functionality is organized into distinct modules, allowing for focused development, debugging, and future enhancement. Each module plays a specific role and interacts cohesively to render a seamless quote-sharing experience. The application consists of the following major modules:

- **User Interface (UI) Module**
- **MainActivity**
- **QuoteAdapter (Custom Adapter Module)**
- **Quote Data Module**

4.1 UI Module (XML Layout)

The UI module defines the layout structure using XML in `activity_main.xml` and `activity_quotes.xml`. This module is responsible for designing the screen elements and organizing them using `LinearLayout`, `ConstraintLayout`, and `RecyclerView`.

4.1.1 Layout Design Using `LinearLayout` and `RecyclerView`

- `activity_main.xml` includes a `TextView` for the daily quote and a `Button` to navigate to the Quotes Page.
- `activity_quotes.xml` uses a `RecyclerView` for listing all the stored quotes in a scrollable list.

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
```

```
    android:orientation="vertical"
```

```
    android:gravity="center"
```

```
    android:layout_width="match_parent"
```

```
    android:layout_height="match_parent">
```

```
<TextView

    android:id="@+id/textQuote"

    android:textSize="18sp"

    android:textStyle="italic"

    android:padding="20dp"

    android:layout_width="wrap_content"

    android:layout_height="wrap_content" />
```

```
<Button

    android:id="@+id/btnNext"

    android:text="New Quote"

    android:layout_width="wrap_content"

    android:layout_height="wrap_content"/>
```

```
<Button

    android:id="@+id/btnViewAll"

    android:text="View All Quotes"

    android:layout_width="wrap_content"

    android:layout_height="wrap_content"/>
```

</LinearLayout>

4.1.2 Responsiveness and Style Considerations

- TextView uses wrap_content and padding for flexibility and visibility.
- RecyclerView ensures efficient rendering of multiple quotes.
- All views adapt to different screen sizes with standard Material Design principles.

4.2 MainActivity

The MainActivity module serves as the app's entry point and controller for the home screen. It is responsible for:

- Setting the content view
- Loading a random quote
- Managing button interactions
- Navigating to the Quotes Page

4.2.1 Lifecycle Methods

The onCreate() method initializes the view, binds UI elements, and sets listeners:

@Override

```
protected void onCreate(Bundle savedInstanceState) {
```

```
    super.onCreate(savedInstanceState);
```

```
    setContentView(R.layout.activity_main);
```

```
    TextView quoteText = findViewById(R.id.textQuote);
```

```
Button btnNext = findViewById(R.id.btnNext);

Button btnViewAll = findViewById(R.id.btnViewAll);


// Display a random quote

quoteText.setText(getRandomQuote());


// On click, display another random quote

btnNext.setOnClickListener(v -> quoteText.setText(getRandomQuote()));


// Navigate to quotes list

btnViewAll.setOnClickListener(v -> {

    Intent intent = new Intent(MainActivity.this, QuotesActivity.class);

    startActivity(intent);

});

}
```

4.3 QuoteAdapter (Custom Adapter Module)

The QuoteAdapter class extends RecyclerView.Adapter and connects the quote data to the RecyclerView.

4.3.1 ViewHolder Binding

The onBindViewHolder() method binds each quote to its layout view:

4.4 Quote Data Module

This module handles the collection of quotes used in the app.

4.4.1 Quote Source Initialization

The quotes are stored as a `String[]` or `ArrayList<String>` in a static class or within the activity:

```
private final String[] quotes = {  
  
    "Believe you can and you're halfway there.",  
  
    "You are stronger than you think.",  
  
    "Do something today that your future self will thank you for.",  
  
    "The best time for new beginnings is now."  
  
};
```

4.4.2 Random Quote Generation

```
private String getRandomQuote() {  
    int index = new Random().nextInt(quotes.length);  
    return quotes[index];  
}
```

4.5 Modular Interaction Flow

Here's a breakdown of how the modules interact:

- UI Module provides the layout and visual structure.
- MainActivity controls the flow of the home screen and navigates to the Quotes Page.
- QuoteAdapter dynamically binds quote data to the list items in `RecyclerView`.
- Quote Data Module stores and supplies quotes to both screens.

This modular separation allows for:

- Easy debugging and future expansion (e.g., adding quote categories or sharing options)
- Clean architecture adhering to best Android development practices
- Scalable design that can incorporate a database or API integration later.

CHAPTER 5

5. IMPLEMENTATION AND RESULTS

The implementation phase of the Daily Quotes Generator Application involved transforming the conceptual design into a functional Android application using Android Studio. The development process included XML-based UI layout design, Java programming for control logic, and integration of text resources representing daily quotes. The outcome is a responsive and lightweight quotes application that displays motivational quotes and allows users to explore all quotes with smooth navigation and minimal delay.

5.1 Development Environment Setup

- IDE Used: Android Studio (Latest Stable Version)
- Programming Language: Java
- Layout Design: XML
- Android SDK Version: API Level 21 (Lollipop) and above
- Minimum SDK Requirement: Android 5.0 (Lollipop)
- Testing Devices: Android Emulator and real devices (OnePlus, Samsung Galaxy)

5.2 Activity and UI Initialization

The MainActivity.java serves as the launching screen of the application. It controls the quote display and navigational logic. In the onCreate() lifecycle method, the layout defined in activity_main.xml is linked and initialized:

```
setContentView(R.layout.activity_main);
```

UI components such as TextView and Button are bound and set with listeners to display a random quote and redirect users to the full list of quotes.

5.3 Dynamic Image Loading with Adapter

The core feature—daily quote rendering—was implemented using a String[] array of motivational quotes. A random index is generated to pick and display a quote each time the app is launched or the "New Quote" button is clicked:

```
private String getRandomQuote() {  
  
    int index = new Random().nextInt(quotes.length);  
  
    return quotes[index];  
  
}
```

This logic is triggered within a `setOnClickListener()` to ensure interactive quote refreshing.

5.4 Quotes Listing with RecyclerView Adapter

The `QuotesActivity.java` uses a `RecyclerView` to display all the stored quotes in a scrollable list. This was implemented with a custom adapter class, `QuoteAdapter`, extending `RecyclerView.Adapter`.

Each item in the list is populated using a `ViewHolder` and a layout file `quote_item.xml`, containing a single `TextView` element for clean, readable quote presentation.

`@Override`

```
public void onBindViewHolder(@NonNull QuoteViewHolder holder, int position) {  
  
    holder.textViewQuote.setText(quoteList.get(position));  
  
}
```

This approach ensures performance efficiency through view recycling and provides scalability for future enhancements like categories or favorites.

5.5 Navigation and Interaction Handling

Intent-based navigation was used for transitioning between `MainActivity` and `QuotesActivity`. This modular navigation structure supports clean code separation and easy maintenance:

```
Intent intent = new Intent(MainActivity.this, QuotesActivity.class);  
  
startActivity(intent);
```

User interaction is kept simple and intuitive, involving minimal clicks and an uncluttered interface.

5.8 Visual Output and Screenshots

The output of the application was verified using both the Android Emulator and real Android devices (e.g., OnePlus Nord 2, Samsung Galaxy series). The following behaviors were confirmed:

- **Accurate Random Quote Display**
- **Responsive User Interface**
- **Successful Navigation.**
- **Error-Free Execution:**
- **Compatibility**

These verifications confirm that the Daily Quotes Generator app performs its intended function reliably and delivers a user-friendly experience.

XML CODE

```
<!-- res/layout/activity_main.xml -->
```

```
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
```

```
    xmlns:tools="http://schemas.android.com/tools"
```

```
    android:layout_width="match_parent"
```

```
    android:layout_height="match_parent"
```

```
    tools:context=".MainActivity">
```

```
    <GridView
```

```
        android:id="@+id/gridView"
```

```
        android:layout_width="match_parent"
```

```
        android:layout_height="match_parent"
```

```
        android:columnWidth="100dp"

        android:numColumns="auto_fit"

        android:verticalSpacing="5dp"

        android:horizontalSpacing="5dp"

        android:gravity="center"/>

</RelativeLayout>
```

JAVA CODE

```
package com.example.quote

import android.os.Bundle

import androidx.activity.ComponentActivity
import androidx.activity.compose.setContent
import androidx.compose.material3.*
import androidx.compose.runtime.*
import androidx.navigation.compose.*

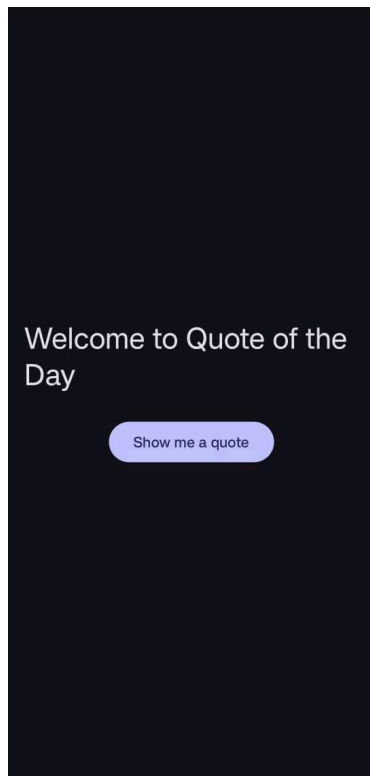
import com.example.quote.ui.theme.QuoteTheme

class MainActivity : ComponentActivity() {

    override fun onCreate(savedInstanceState: Bundle?) {

        super.onCreate(savedInstanceState)
```

```
setContent {  
  
    QuoteTheme {  
  
        val navController = rememberNavController()  
  
        NavHost(navController, startDestination = "home") {  
  
            composable("home") { HomeScreen(navController) }  
  
            composable("quote") { QuoteScreen() }  
  
        }  
  
    }  
  
}  
  
}
```



CHAPTER 7

7.CONCLUSION

The Daily Quotes Generator application successfully fulfills its primary objective of delivering motivational and inspirational quotes through a simple and user-friendly Android interface. The application offers two main functionalities: displaying a randomly selected quote on the home screen and allowing users to browse through all available quotes on a dedicated screen.

By employing a modular architecture with clear separation of concerns, the app ensures maintainability, scalability, and performance optimization. The use of native Android components such as Activity, Intent, and RecyclerView facilitates efficient rendering and navigation, while keeping the application lightweight and responsive.

Through thorough testing on both emulators and real Android devices, the app has demonstrated stable performance, accurate quote delivery, and smooth user interactions. The clean UI and offline functionality make it a reliable tool for users seeking daily motivation without the need for internet connectivity.

Overall, the application stands as a functional and extendable solution that can be further enhanced with features such as quote categorization, daily notifications, or cloud-based updates in future iterations.

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